

[6354]-625

B.E. (Mechanical Engineering)

COMPUTER INTEGRATED MANUFACTURING

(2019 Pattern) (Semester - VIII) (402048)

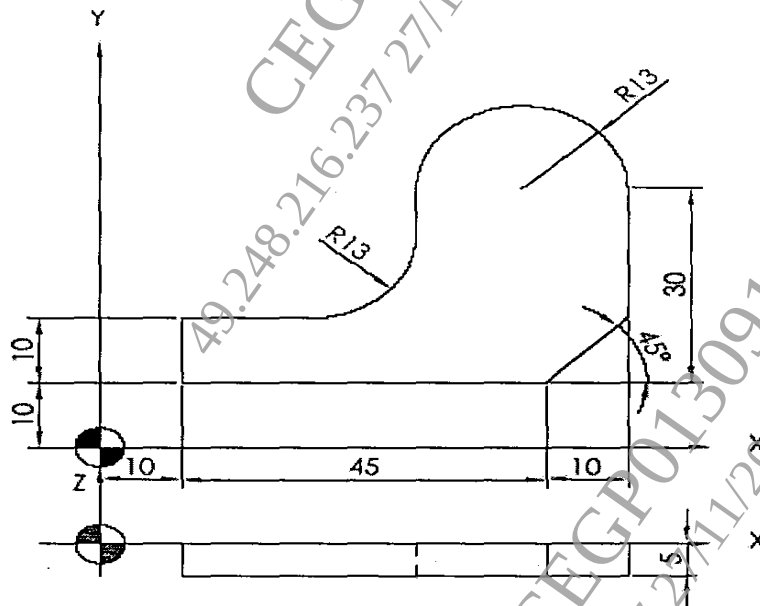
Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1)** a) Explain and illustrate coordinate System used in CNC Turning Center. [6]
 b) Write a CNC program for the part shown in Figure 1b using an end mill of 6mm diameter. Assume suitable data for machining parameters. Raw material size – 75mm × 75mm × 5mm. [12]



(All dimensions are in mm)

Fig. 1b

OR

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- Q2) a)** Explain and illustrate three Canned Cycles with suitable examples. [6]
b) Write a complete part program using G and M codes for the job shown in Figure 2b. The material of the job is MS. Assume suitable speed and feed for machining. Billet size: diameter = 60mm and length = 80 mm, Thread: Major Diameter: 16mm, and Pitch = 1.5 mm, Groove = 5mm and depth = 2.5mm. Use suitable Roughing and Finishing Cycles and appropriate Canned Cycles. [12]

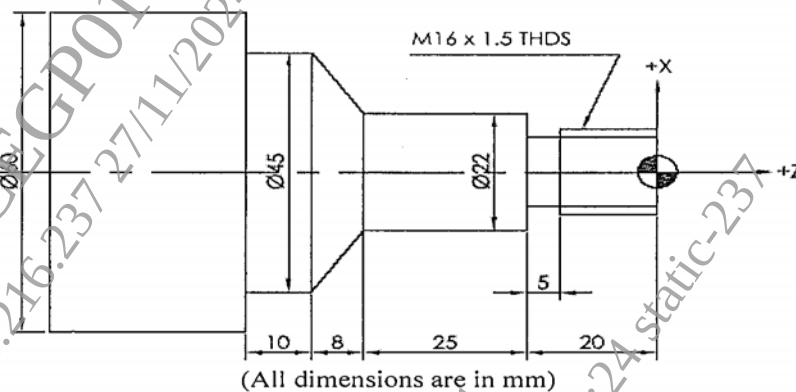


Fig. 2b

- Q3) a)** Explain working and advantages of Enterprise Resource Planning (ERP). [9]
b) Explain approaches of Computer Aided Process Planning with their Advantages and Disadvantages. [8]

OR

- Q4) a)** Draw and Explain Computer Aided Inspection and Quality Control System. [9]
b) What is Manufacturing Resource Planning MRP II? Explain with diagram. [8]

- Q5) a)** Explain and illustrate four layouts in FMS systems. [10]
b) Discuss any four types of material handling systems used in industry. [8]

OR

- Q6) a)** Explain and illustrate the concept of composite part. [9]
b) Write applications of Rank Order Clustering Method (ROCM) in Mechanical Industry. [9]

- Q7) a)** Explain the classification of components of Industry 4.0. [8]
b) Explain and illustrate Cyber-Physical Manufacturing Systems. [9]

OR

- Q8) a)** Explain with an example, How Digital Twin is implemented for Smart Manufacturing? [9]
b) Compare Industry 4.0 with Industry 5.0. [8]

